

Input Ontology (IO)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: MILU | Context: North Indian Hindi-Medium Postgraduate Teacher — MCQ Evaluation Deployment

Input Ontology definition: Whether the benchmark’s test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MILU’s taxonomy of 8 domains and 41 subjects spans the breadth needed for a multi-subject teacher-evaluation deployment, and the format is exclusively single-correct-answer MCQ, which is a direct structural match for the deployment’s binary positive/negative marking requirement. However, the taxonomy is derived from competitive exam content (UPSC/SSC/state PSC) rather than school-board syllabi, and the dataset sample shows a heavy skew toward Engineering & Tech (23% of Hindi validation items) with under-representation of Civics/Political Science and Panchayati Raj content that is central to UP/MP/Rajasthan/Bihar state-board curricula. Per-subject statistics for Hindi specifically are not publicly available, making it impossible to verify whether the categories adequately cover state-board curricular framing.

Strengths

- Exclusive single-answer MCQ format directly matches the deployment’s binary scoring requirement [Q22, Q33, DATASET-D1]
- All 8 declared domains appear in the Hindi validation sample, providing nominal multi-domain breadth [DATASET-D1, D2, D4, D9, D20]
- Indic LLM sub-taxonomy is treated separately, indicating awareness of system heterogeneity [Q60]

ID	Evidence
Q22	“MILU is a large, multi-domain test set containing multiple-choice based questions (MCQs) taken from over 41 subjects with an emphasis on India-specific knowledge.” (p.3)
Q33	“During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency.” (p.4)
D1	जब एक डीसी सीरीज मोटर बर्ना लोड के चलती है: ... मोटर की गति बहुत अधिक होती है — Engineering question about DC series motor — translated from English
D2	नमिनलखिति में से कौन सा खनजि उत्तर प्रदेश में नहीं पाया जाता है? ... अभ्रक — North Indian state-specific geography: UP minerals question
D4	राष्ट्रीय आपातकाल घोषित करने के लिए 'सशस्त्र विद्रोह' शब्द संविधान में कब जोड़ा गया? 44वें संविधान संशोधन अधिनियम द्वारा — Constitutional law MCQ — Law & Governance domain
D9	नमिनलखिति में से कौन प्रत्यक्ष कर का उदाहरण है? संपत्तिकर — Economics/taxation MCQ — pan-India competitive exam register
D20	मई 2022 में इराक के सुलेमानिया में आयोजित तीरंदाजी एशिया कप 2022 स्टेज 2 अभियान में भारत ने कितने स्वर्ण पदक जीते? आठ — Sports GK — India-centric, recent current affairs
Q60	“We evaluate around 16 Indic language LLMs on MILU. These models are primarily built by adapting English LLMs, such as LLAMA-2-7B, by first continually pretraining on small amount Indic language data, followed by optionally instruction finetuning them.” (p.6)

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment

IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories for the deployment include Hindi Literature (canonical board authors), History (regional medieval and freedom movement), Civics/Political Science with Panchayati Raj and state administration, Geography of North India, Science/Maths in Hindi, Economics, and General Knowledge. MILU declares all 8 domains and 41 subjects [Q42] which nominally cover this breadth, with subject-level statistics in supplementary Table 9 [Q98].

IO-2: MILU's taxonomy is derived from competitive exam content [Q23, Q24, Q27] rather than school/university board syllabi. State-specific civics (Panchayati Raj, state legislatures) is not visible as a dedicated category in the sample [DATASET-D4 is the only Law & Governance item, and concerns the 44th constitutional amendment, not state-level civics]. The single dedicated Hindi-board literature register (close reading of Tulsidas/Premchand/Kabir) appears absent from the sample [DATASET-D5 — the only Literature item is about Wordsworth].

IO-3: Engineering & Tech dominates the Hindi validation sample (6 of 26 items) including DC motors, transformers, rectifiers, and FORTRAN [DATASET-D1, D13, D14, D23], which are characteristic of polytechnic/engineering entrance content rather than secondary or degree-college teacher domains. This represents construct-irrelevant variance for a general Hindi-medium teacher deployment.

IO-4: Identified gaps: (a) school-literature framing for Hindi canonical authors absent; (b) state-specific civics/Panchayati Raj under-represented; (c) over-representation of engineering content; (d) per-language per-subject distributions not published [NOT FOUND per regional YAML state_wise_exam_source_distribution_for_hindi].

ID	Evidence
Q42	"Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies." (p.4)
Q98	"Table 9: Detailed subject level statistics of MILU across different languages." (p.20)
Q23	"This benchmark covers many domains, including Science, Social Sciences, Humanities, Arts, Business Studies, and Law, among others." (p.3)
Q24	"These questions were sourced following an approach similar to AGIEVAL (Zhong et al., 2023), collecting the questions from various public exams taken by individuals intending to either pursue higher studies or seek career advancements, such as qualification tests and national and state-level civil services exams, among others." (p.3)
Q27	"Our benchmark includes questions from over 40 different types of exams conducted both at the national and state levels over recent years." (p.3)
D4	राष्ट्रीय आपातकाल घोषित करने के लिए 'सशस्त्र विद्रोह' शब्द संविधान में कब जोड़ा गया? 44वें संविधान संशोधन अधिनियम द्वारा — Constitutional law MCQ — Law & Governance domain
D5	विलियम वर्ड्सवर्थ _____ के कवि हैं। इंग्लैंड — Literature question about English poet Wordsworth — not Hindi-board canonical literature
D1	जब एक डीसी सीरीज मोटर बर्ना लोड के चलती है: ... मोटर की गति बहुत अधिक होती है — Engineering question about DC series motor — translated from English
D13	हाफ वेव रेक्टिफायर का आउटपुट क्या होता है: पल्सेटिंग डीसी — Electrical engineering MCQ — translated from English
D14	स्थिर धारा ट्रांसफार्मर _____ प्रकार का होता है। शेल — Engineering transformer question — translated item with 'शेल' (shell) transliteration
D23	फॉरट्रान 77 के फक्सिड फॉर्मेट में कॉलम 2 से 5 में संख्या का क्या उद्देश्य होता है? एक जंप लेबल या फॉर्मेट लेबल — Computer science MCQ — translated, technical vocabulary
WEB-9	HuggingFace dataset is gated, limiting independent verification of per-subject Hindi distributions (https://huggingface.co/datasets/ai4bharat/MILU)
WEB-12	GitHub subject categories include 'Literature and Linguistics' and 'Arts and Culture' but no item-level framing detail published (https://github.com/AI4Bharat/MILU)

Information Gaps

- Per-language per-subject item counts for Hindi not published in accessible form (Table 9 in gated PDF)
- Whether the test split's domain distribution matches the sampled validation distribution
- Whether MILU includes any items framed in school-literature register vs. competitive-exam GK

Requires Expert Verification

- Whether the 41-subject taxonomy maps cleanly onto UP/MP/Rajasthan/Bihar board syllabi at the subject level
 - Whether the missing categories (state civics, Hindi literary close reading) are material to teacher use cases or marginal
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Remediation

Gap 1: Heavy skew toward Engineering & Tech (23% of Hindi sample) and under-representation of Civics/Political Science, Panchayati Raj, and state-board literature framings

Recommendation: Use MILU domain-level results as a coarse signal only; build a deployment-specific subject inventory that mirrors UP/MP/Rajasthan/Bihar board syllabi, and weight or filter MILU subjects to align with actual teacher use cases